JADY PAMELLA

AI Engineer focused on AI Safety | PhD Fellow, Aarhus University | Elsevier 2025 | EA Sweden Fellow 2026 | BlueDot Impact

AI Engineer focused on AI safety who builds trustworthy production AI. 12 years shipping AI inside regulated environments (a public-sector bank, federal research labs, and climate tech), from LLM platforms and RAG pipelines to computer vision and MLOps at safety-critical latency budgets. PhD at Aarhus University on digital twins for district cooling (SWiM, funded by the Grundfos Foundation) adds depth in AI safety, cybersecurity, and critical-infrastructure resilience. Peer-reviewed in Elsevier Computers and Security.

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Fellowships

EA Sweden Fellowship
Effective Altruism Sweden · 2026 cohort

AGI Strategy + Technical AI Safety
BlueDot Impact · cohorts started 2026 · Future of AI completed 2025

Network for Global Professionals
Swedish Institute · 2024 to 2026

Education

Stockholm University, DSV
MSc Computer and Systems Sciences (AI and IoT). Aug 2024 to May 2026. Swedish Institute Scholar (SISGP).

University of Brasilia, PPEE
MEng Electrical Engineering, concentration in Cybersecurity and Risk Management. Mar 2024 to May 2026.

UPIS Faculdades Integradas
BSc Information Systems. 2010 to 2013, Brasilia.

Volunteer Work

Student Ambassador
Drivhuset Stockholm · 2024 to 2026

Open-source contributor
Stiftelsen Effekt / Ge Effektivt · geeffektivt.se · 2026

Web Developer
Alla Ska Få Mat, Stockholm · 2024 to present

Publications

Alves, R. S., **Barbacena da Silva, J. P.**, Ribeiro Jr., L. A., Rabelo Nunes, R. (2025). *Enhancing Cybersecurity in the Judiciary: Integrating Additional Controls into the CIS Framework.* Computers and Security (Elsevier), IF 5.6. DOI 10.1016/j.cose.2025.104584.

Barbacena da Silva, J. P., et al. (2024). *Cybersecurity Protection in the Brazilian Judiciary.* NAVUS, Journal of Technology and Management. DOI 10.22279/navus.v14.2032.

Barbacena da Silva, J. P., et al. (2025). *Critical Infrastructure Under Threat: Business Risks for Cybersecurity in Metropolitan Transport Systems.* ENANGRAD 2025 (Brazilian National Meeting of Administration Programs).

Languages

- Portuguese (native)
- English (TOEFL 101/120)
- Spanish (B1)
- Swedish (SFI C)

Technical Stack

- Python, TypeScript, SQL, PyTorch, TensorFlow
- LLMs, RAG, agents, evals, structured outputs
- AWS, GCP, Kubernetes, Docker, Terraform
- Supabase, PostgreSQL, Spark
- COBIT, ITIL, CIS Controls, ISO 27001, GDPR

Experience

Aarhus University – PhD Fellow, Digital Twins for District Cooling (SWiM)
Aug 2026 - Present (Aarhus)

Digital twins for district cooling in the SWiM project, at the intersection of AI safety, cybersecurity, and critical-infrastructure resilience.

- Verifying that **district cooling installations are commissioned as they were designed**, combining Building Information Modelling, AI and digital twins. Part of SWiM, a five-year project on sustainable water-based cooling for megacities funded by the Grundfos Foundation, supervised by Peter Gorm Larsen and Carl Schultz.

Deep Forestry AB – AI Research Engineer
Dec 2025 - Jun 2026 (Uppsala)

MSc thesis on safe landing-zone detection for autonomous drones in below-canopy forests.

- Built a hybrid geometric and neural pipeline (RANSAC, PointNet, PointNet++, DGCNN) reaching **F1 0.926 at 53 ms per frame on NVIDIA Jetson Orin NX**, a **29× latency reduction (5.8s to 200ms)** over the deep-learning baseline.
- Combined KD-Tree indexing and voxel downsampling with the neural stack to hit the edge latency budget. Counterfactual gain validated with session-level cross-validation on real LiDAR flight data.

SoiQet – Founder and CEO
Aug 2025 - Present (Aarhus)

AI-powered social media platform for creators, brands, and teams (Plug Into Social Intelligence), from prototype to production.

- Applied AI-engineering practices across **6 LLM providers** (OpenAI, Anthropic, and others), including model evaluation, so model-swap and build-vs-buy decisions are measurable (output stability and structured-output adherence) instead of guesswork.
- Shipped production in-house RAG with structured outputs, guardrails, OAuth across 6 platforms, Stripe billing, and Supabase row-level security. Chose RLS over Firebase because B2B multi-tenant access boundaries had to be auditable.
- Assembled and led the team, set the roadmap, and own delivery from build to launch.

Independent – Technical Advisor (AI-powered consumer product)
2026 (Remote)

- Delivered a **60-page technical audit** of an AI-powered consumer product MVP. Identified **20 of 21 Supabase tables without RLS** (verified exposure with public API key), **29 missing AI and security guardrails**, and AI cost runaway risk. Delivered product strategy reframe plus 8-week prioritised roadmap.

LabRisk, University of Brasilia – Cybersecurity Research Engineer
Aug 2024 - Present (Brasilia)

- Research on risk management, AI for critical-infrastructure cybersecurity, and security governance frameworks. Output includes the 2025 Elsevier *Computers and Security* paper extending the CIS framework for the Brazilian judiciary, and the 2025 ENANGRAD conference paper on cybersecurity business risks in metropolitan transport systems.

Bank of Brasilia – BRB – IT Analyst and DevOps Engineer
Dec 2014 - Present (Brasilia)

- Delivered a production NLP chatbot (Python, TensorFlow) that **cut response time 40% for 100K+ users**, then led the build-vs-buy pivot that routes **40% of BRB customer service through the resulting WhatsApp channel**. Built GDPR-aligned analytics for the instant payments system on PySpark.
- Rebuilt the release pipeline on Jenkins, GitLab, Docker, Kubernetes, OpenShift. **Deployment time cut 60%**. Held the rollout for 2 months to close the compliance gaps first, in a regulated banking environment.

Bank of Brasilia – BRB – IT Manager and Engineering Manager
May 2016 - Jul 2024 (Brasilia)

- Owned a **$1M+ annual IT budget** and a **team of 10+ engineers**. Cut operating cost **15%**, saved **$150K+ in vendor renegotiation**, lifted audit accuracy **40%** in a regulated banking environment.

Tree++ – Co-founder and Head of Innovation
Oct 2024 - Present (Stockholm)

- Climate-tech non-profit. Built the core **computer-vision pipeline and ML matchmaking system** from zero to production, enabling 1,000+ trees planted across 10 countries.

Awards and Recognition

- **1st place, EpiMinds Google AI Hackathon (2026).** Shipped a Multi-Agent Crisis Response System (MACS) in a single day.
- **1st place, Kong and Kolomolo AI-Assisted Coding Hackathons (2025).**
- **Judge, Apart Research Global South AI Safety Hackathon (2026).** Technical AI Safety, Governance and Security tracks.
- **Speaker, Nordic Tech Week "New Tech, Old Rules" panel (2026).**